AIRFOIL IMPORTANCE FOR PROPELLER OPTIMIZED DESIGN

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ABSTRACT

Nowadays, most aerial low-emission propulsion systems are based on propellers. This includes the majority of UAM (urban air mobility) configurations, which are eVTOL (electric vertical take-off and landing) equipped with propeller-based electric propulsion systems. Hence, optimized propeller design is crucial for the vehicle capabilities. An optimized propeller might reduce dramatically the vehicle emissions, improves its acoustic signature, and increases its performance.

Similarly, optimized airfoil design influences its wing characteristics. Wrong airfoil might cause a degraded performance to a winged air-vehicle or even unacceptable aerodynamic behaviour. In many aspects, propeller might be referred as a “rotating wing”, with its rotated blades operating as a wing. Thus, one might consider the same importance of airfoil upon propeller performance and its aerodynamic properties.

These two paradigms can be summed as:

- Optimized propeller design is essential to vehicle characteristics
- Optimized airfoil design is essential to propeller characteristics

The outcome of these two saying, are numerous researches dealing with airfoil optimal design for propellers. The current paper tries to inspect the importance for such efforts by presenting the impact of air-section’s 2-D aerodynamic characteristics on propeller performance.

The paper includes an analysis which shows that airfoil’s 2-D sectional lift-to-drag ratio is the prominent property influencing propeller performance. The analysis also proves that airfoil lift-to-drag ratio influence is small on the propeller performance.

Airfoil drag is only one of 3 contributors for propeller’s required-power. Most of the required-power is due to the propulsive-power and induced-power. Thus, airfoil drag has low influence on the total required-power.

It is obvious that lift-to-drag ratio per-se is not the only criteria for airfoil selection. The paper discusses all other airfoil considerations, concerning rotating wings and gives some guidelines for choosing proper cross-sections for propeller’s blade.

The paper proves that as-long-as the airfoil has reasonable drag characteristics, no effort should be invested with improving its 2-D aerodynamic properties. Hence, using optimization to design propeller by geometric shaping of its air-sections might be irrelevant. One should not consider investing high resources on such minor design feature. For propellers, it is much beneficial to choose an existing air-section and perform optimization mainly on the blade’s chord and twist distribution, without modifying the air-section geometry.

1. INTRODUCTION

Propellers are rotating wing in nature. This lesson was learned early in the flight history and implemented by the Wright brothers [1]. As such, every wing cross section seems to influence its characteristics. Hence, designing a wing, starts with careful choice of its air-section. This is not the case with propellers, as will be shown in the current effort. Literature provide various examples to this claim.

Moffitt et-al [2] discussed thoroughly the importance of inaccuracies of various propeller’s parameters upon its performance. In their analysis, the authors show that the propagation of the airfoil drag coefficient to the propeller integral performance is very small.

Reuben et-al [3] refer to the importance of airfoil at propellers, mainly at low Reynolds number. Their finds shed light on the 2-D conditions which these airfoils are operated. This emphasizes the importance of high lift--to-drag properties, rather high lift capabilities.

Traub [4] expanded the well-known Larabee’s [5] formulation of optimal designed propeller method. He introduced Reynolds-number dependency of the air-sections’ 2-D aerodynamic characteristics. The results showed that

> *achieving a peak local radial lift-to-drag ratio along the blade does not appear to have a significant impact on the net efficiency of the propeller*

Thus, the 2-D improvement of lift-to-drag ratio is less important than the induced characteristics of the propeller.

The importance of airfoil was demonstrated also for hovering-propeller, i.e. rotor. Vu et-al [6] conducted numeric optimization of rotor blade at hovering flight condition. They started with standard NACA-0012 and constant chord blade. The optimized design exhibited small airfoil modifications, with radically altered chord distribution - high tapered ratio blade.

There are places where airfoils are important such as in the case of Bingham [7] paper. He evaluated airfoils for helicopter rotor, focusing mainly on their drag-divergence performance. Hence, mainly at off-design conditions of the blades' tip. At design conditions, the airfoils impact is quite small.

In a case of different "rotating wing" usage, Kanya and Visser [8] discussed the importance of airfoil for horizontal axis wind turbine. In their research they found that the penalty for using common airfoil such as 4-digit NACA family over specialized airfoil is quite small.

In that sense, the current effort will concentrate on propeller performance influenced by airfoil drag (or drag-to-lift ratio). This is, as will be shown later, the major aerodynamic influence of the blade's air-section over the propeller performance.

Note that in addition to drag, there are many additional considerations for airfoil selection. These can be summarized in the following list:

a. Low pitching-moment. This is extremely important in helicopter rotors. For rotors, high pitching moment can cause extreme fatigue stresses on the pitch-links and servos. This is the reason why rotor airfoils do not exhibit high camber, e.g. Ref. [9] which presents the UH-60 airfoil. Reference [10] takes this airfoil and tries to improve it using optimization which maintains its pitching moment characteristics with decreased drag. The results emphasize the small benefit of decreased drag with pitching moment penalty.
b. High thickness due to structural considerations. This is mostly important for wind turbines blades [11] but also for root cross-sections in propellers and rotors. Structural properties of rotating wings cannot be compromised; thus, thickness constraints are rigid as part of the design process.
c. Trailing edge thickness due to manufacturing constraints. For composite/natural-composite propellers, the trailing edge thickness, especially close to the blade tip might constraint significantly the airfoil choice. This is more pronounce as the propeller diameter and chords decrease.
d. High critical and drag-divergence Mach number characteristics. This requirement is important for high speed propellers or high rotational-speed/large radius blades. [7] For hovering propeller or small-size propellers this is somewhat less important. Still, there are some exotic cases such as the Mars helicopter which this requirement is crucial. [12]
e. Stall characteristics and maximum lift-coefficient is not a major requirement for propeller's air-sections. Although high lift-coefficient might be required at off-design flight regimes. In most cases, the design conditions try to exhibit moderate lift coefficient, in which the lift-to-drag ratio is optimal. Moreover, in propellers, the stall characteristics are very different from the 2-D characteristics due to the blade rotation. In most cases, stall is delayed to much higher angles-of-attack. [13]

Saying that, the current effort emphasizes the propeller performance and its dependency on the cross sectional drag. This explains the reason for the relatively lack-of-importance considering airfoil choice for propellers. Some numeric examples will demonstrate the impact of "good" versus "excellent" airfoil choice and its influence on the overall propeller performance.

2. PROPELLER THRUST AND POWER

To estimate the importance of propeller's air-sections one should refer to the thrust and required power. Propeller is used to produce thrust, T. The penalty for this thrust is the required power, P, as invested in the propeller shaft. Propeller performance can be characterized as the amount of P for required T or the amount of produced T for available P.

In what follows, only axial flight is considered, hence flight velocity, V, coincides with the propeller rotation axis – the shaft. For this flight conditions, required power is consisted of three contributions:

a. Propulsive thrust power
b. Induced power
c. Cross-sectional drag power

Propulsive thrust power if the product of thrust to flight velocity, $T \cdot V$. The induced power is dependent on the induced velocity, v_a, which the propeller produces while accelerating the air. This gives an induced power which is the product of thrust with induced velocity, $T \cdot v_a$.

Here a constant induced-velocity on the propeller disk is considered. This is the famous Betz's condition [14] for optimal propeller design. In actual propellers, the designer tries to reach such homogeneous induced-velocity distribution. Thus, in most cases, propeller exhibits rather constant induced-velocity at axial operation.

Cross-sectional drag power, P_{Cd}, represents the power required to overcome the parasite 2-D drag of the blade's cross-sections. This drag is represented through the cross-sectional drag coefficient, C_d.

Total required power, P, can be written as the sum of the above contributions as depicted in Eq. (1).

$$P = T \cdot (V + v_a) + P_{Cd} \tag{1}$$

The propeller efficiency, η, is defined as the ratio between the propulsive thrust power to the total required power:

$$\eta = \frac{T \cdot V}{P} = \frac{1}{1 + \frac{v_a}{V} + \frac{P_{Cd}}{T \cdot V}} \tag{2}$$

Note that for static operation, V=0, the efficiency equals zero. For hover-propeller substitute parameter is defined as a Figure-of-Merit, *FM*. It is defined as the ratio between the ideal induced-power to total power, which is somewhat different from efficiency definition:

$$FM = \frac{T \cdot v_{a,ideal}}{P} = \frac{T \cdot v_{a,ideal}}{T \cdot v_a + P_{Cd}} \tag{3}$$

These two metrics, η and *FM*, will be used separately for axial-flight and hovering propeller, respectively. These terms are to be simplified, thus enabling estimation of the airfoil aerodynamic characteristics impact.

The ideal axial induced velocity, $v_{a,ideal}$, is given using simple momentum theory as developed by Glauert [15]:

$$v_{a,ideal} = -\frac{V}{2} + \sqrt{\left(\frac{V}{2}\right)^2 + \frac{T}{2 \cdot \rho \cdot A}} \tag{4}$$

Note that $v_{a,ideal}$ represents the minimal induced velocity. This was presented in various occasions [20][21][22], hence Eq. (4) proved to under-predicts the actual axial-induced velocity. This fact can be treated by using a correction factor κ_a >1, which relates between the actual v_a, and $v_{a,ideal}$.

$$v_a = \kappa_a \cdot v_{a,ideal} \tag{5}$$

As shown in Figure 1, the actual induced velocity is higher compared to the ideal. In this figure, the axial flight velocity, V, and axial induced velocity, v_a, are normalized by the ideal hover induced velocity, $v_{a,h,ideal}$.

At normal axial operation of propellers, where V is high, thus $V/v_{a,h,ideal} \geq 1.5$, κ_a is close to unity. As speed decreases, the induced velocity deviates from the axial momentum ideal estimation. At hover the correction factor reaches to κ_a =1.2÷1.3 .

Using Eq. (4) for static condition, V =0, and substituting into *FM* definition in Eq.(3) gives the well-known relation between propeller thrust and required-power at hover:

$$FM = \frac{T^{1.5}}{P \cdot \sqrt{2 \cdot \rho \cdot A}} \tag{6}$$

Cross-sectional drag power can be estimated using an equivalent drag coefficient, C_{d0}, which represents the average 2-D parasite drag of the blade's distributed airfoils. Using blade-element model, BEM, [22] the cross-sectional drag power are defined in eq. (7).

$$\frac{P_{Cd}}{\rho \cdot A \cdot V_{tip}^{\ 3}} = \frac{\sigma \cdot C_{d0}}{8} \tag{7}$$

where σ is the propeller solidity, i.e. the ratio between the blade area to disk area, as defined in Eq. (8).

$$\sigma = \frac{N_b \cdot c_{av} \cdot R}{A} \tag{8}$$

N_b is number-of-blades and c_{av} is the average blade chord.

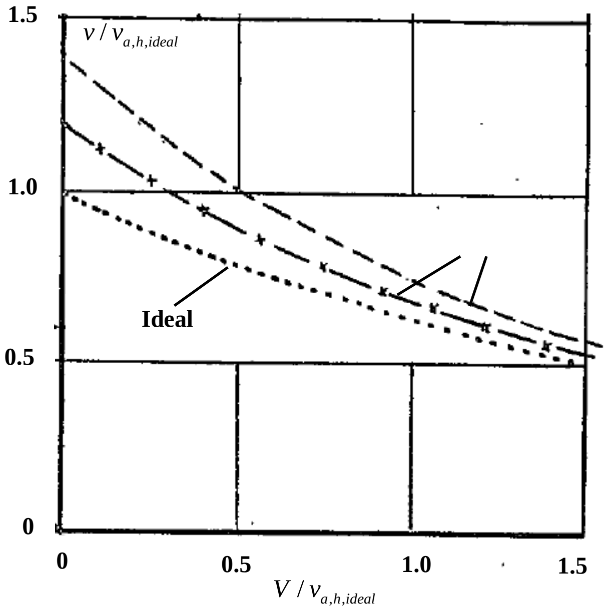

Figure 1, Ideal induced velocity versus experimental results. Taken from [16]

Note that the equivalent drag coefficient, C_{d0}, is not the average of the 2-D drag-coefficient distribution, C_d, but the average weighted by the radial coordinate, $r^3 \cdot C_d$:

$$C_{d0} = \frac{4}{R^4} \cdot \int_0^R C_d(r) \cdot r^3 \cdot dr \tag{9}$$

In propellers, similar measure for the blade-solidity, σ, is the Activity-Factor per blade, *AF*, which is defined, following Eq. (9), with chord weighted by r^3. Note that the factor 100,000/32, in this expression, is required to make *AF's* value at the ball-park of 100.

$$AF = \frac{100{,}000}{32 \cdot R^4} \int_0^R \frac{c}{R}(r) \cdot r^3 \cdot dr \tag{10}$$

In what follows, propeller disk area, A, air density, ρ, and blade tip velocity, V_{tip}, are used as normalization factors. V_{tip} is defined as the tangential velocity of the blade tip at static condition, V=0:

$$V_{tip} = \Omega \cdot R \tag{11}$$

where Ω is the propeller angular velocity and R is the propeller radius. The thrust and power coefficient, C_T and C_P, respectively, are defined using these normalization factors.

$$C_T = \frac{T}{\rho \cdot A \cdot V_{tip}^{\;2}} \qquad C_P = \frac{P}{\rho \cdot A \cdot V_{tip}^{\;3}} \tag{12}$$

In propeller, thrust produced through the cross-sectional lift. Using again BEM and assuming a representative design lift coefficient, C_{l0} :

$$C_T = \frac{\sigma \cdot C_{l0}}{6} \tag{13}$$

Here, similar to C_{d0}, the design lift coefficient, C_{l0}, is defined using r^2 weighting:

$$C_{l0} = \frac{3}{R^3} \cdot \int_0^R C_l(r) \cdot r^2 \cdot dr \tag{14}$$

All of the above relations are now used to simplify the efficiency, η, and figure-of-merit, FM, expressions.

3. η INFLUENCED BY 2-D AERO.

Substituting Eqs. (7) through (13) to Eq. (2) gives an estimation for the propeller efficiency.

$$\eta = \frac{1}{\frac{1}{2}(2-\kappa_a) + \kappa_a \cdot \sqrt{\frac{1}{4} + \frac{\pi^3}{3} \cdot \frac{\sigma \cdot C_{l0}}{J^2}} + \frac{3\pi}{2} \frac{1}{J \cdot C_{l0} / C_{d0}}} \tag{15}$$

where J is the advance ratio defines as:

$$J = 2\pi \frac{V}{V_{tip}} \tag{16}$$

For propellers, as flight speed increases, the induced factor, κ_a, becomes unity. This gives a good approximation for propeller efficiency at design conditions.

$$\eta = \frac{1}{\frac{1}{2} + \sqrt{\frac{1}{4} + \frac{\pi^3}{3} \cdot \frac{\sigma \cdot C_{l0}}{J^2}} + \frac{3\pi}{2} \frac{1}{J \cdot C_{l0} / C_{d0}}} \tag{17}$$

As J increases, η reaches unity. This is true due to the increased part of the propulsive thrust power, thus the induced and cross-sectional drag power become negligible.

The optimum efficiency is considered for minimum lift-to-drag, as shown from the term of C_{l0} / C_{d0}. One should remember that C_{l0} is not arbitrary. The required thrust dictates the design lift-coefficient through the relation in Eq. (13). This makes the design of a propeller a combination of the correct choice of its solidity, σ, which brings the 2-D lift coefficient, C_{l0}, to the proper lift-to-drag region.

Equation (17) suggests that for given geometry, and thrust demand, the change of airfoil is expressed only in the lift-to-drag ratio.

Figure 2 shows the 2-D aerodynamic characteristics for NACA-0012 airfoil. The pick lift-to-drag is at about C_l = 0.8 and it reaches C_l / C_d =75. Here NACA-0012 is used only as an example. One can find much higher lift-to-drag airfoils in the literature [19].

Figure 3 demonstrates the influence of lift-to-drag ratio on propeller efficiency. Even increasing the lift-to-drag ratio to 150, efficiency increases by about 1% compared to the efficiency at C_{l0} / C_{d0} = 75. In addition, compared to the lift-to-drag ratio, advance ratio has much substantial influence.

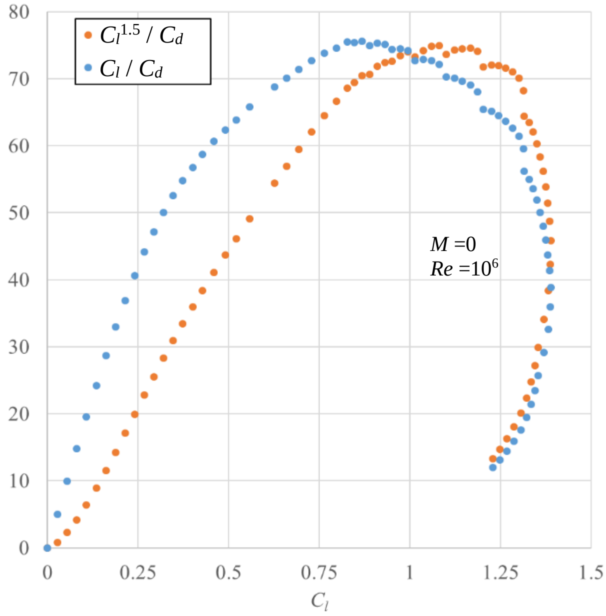

Figure 2, NACA-0012 aerodynamic characteristics, source: http://airfoiltools.com/

4. *FM* INFLUENCED BY 2-D PROPERTIES

Substituting Eqs. (11) through (13) to Eq. (3) gives an estimation for the propeller Figure-of-Merit at static conditions, V =0:

$$FM = \frac{1}{\kappa_a + \frac{3\sqrt{3}}{2} \frac{1}{\sqrt{\sigma}} \frac{1}{C_{l0}^{\;1.5} / C_{d0}}} \tag{18}$$

This result is somewhat more elegant compared to the efficiency result of Eq. (15) due to static condition. Moreover, the 2-D aerodynamic parameter of $C_{l0}^{1.5} / C_{d0}$ is noticeable. As shown in Figure 2, the aerodynamic 2-D parameter reaches $C_l^{1.5} / C_d$ = 75 for the NACA-0012. For perfect blade, without drag, the FM get close to unity, as dictates by the induced factor, κ_a.

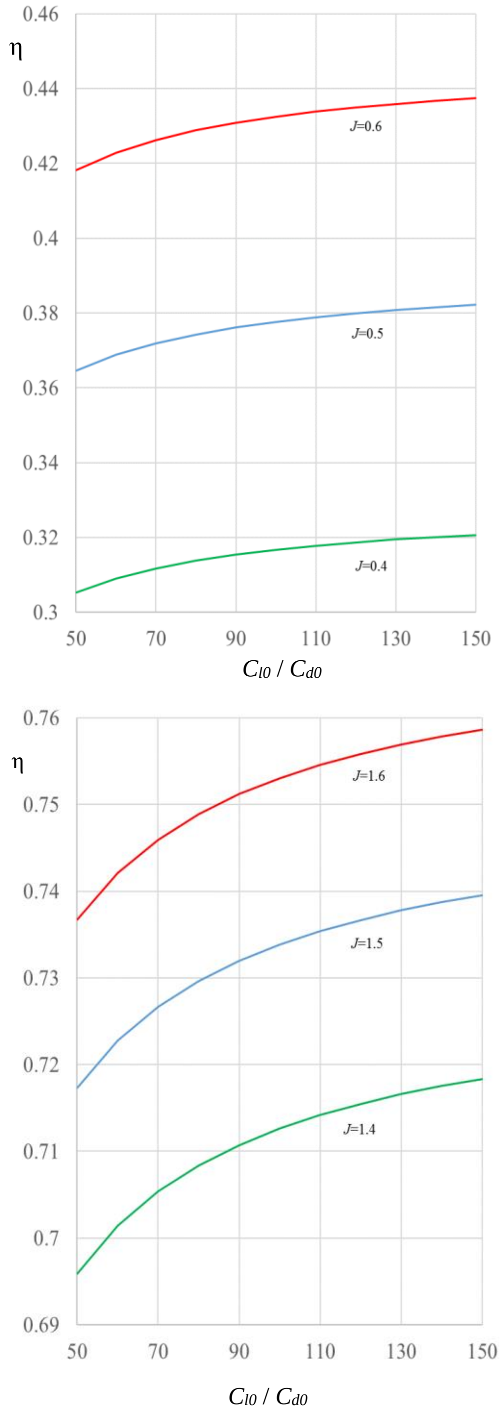

Figure 3, Propeller efficiency estimations, σ=0.04
Note the zoomed vertical scale which allows noticing
C_{l0} / C_{d0} influence

Upper chart of Figure 4 shows the influence of $C_{l0}^{1.5}/C_{d0}$ on *FM*. Although this influence is somewhat higher compared to the efficiency chart of Figure 3, increase or decrease with $C_{l0}^{1.5}$ / C_{d0} influences *FM* in moderate manner. For example, increase $C_{l0}^{1.5}/C_{d0}$=70 to $C_{l0}^{1.5}/C_{d0}$=100 results with 2% *FM* change. As *FM* becomes lower, influence of $C_{l0}^{1.5}/C_{d0}$ increases - still it is moderate.
Better insight is achieved from the lower chart in Figure 4. It shows the ratio between the cross-sectional drag power, P_{Cd}, to the total required power, *P*. One can find the relation between the two charts through Eq. (19).

$$\frac{P_{Cd}}{P} = 1 - FM \cdot \kappa_a \tag{19}$$

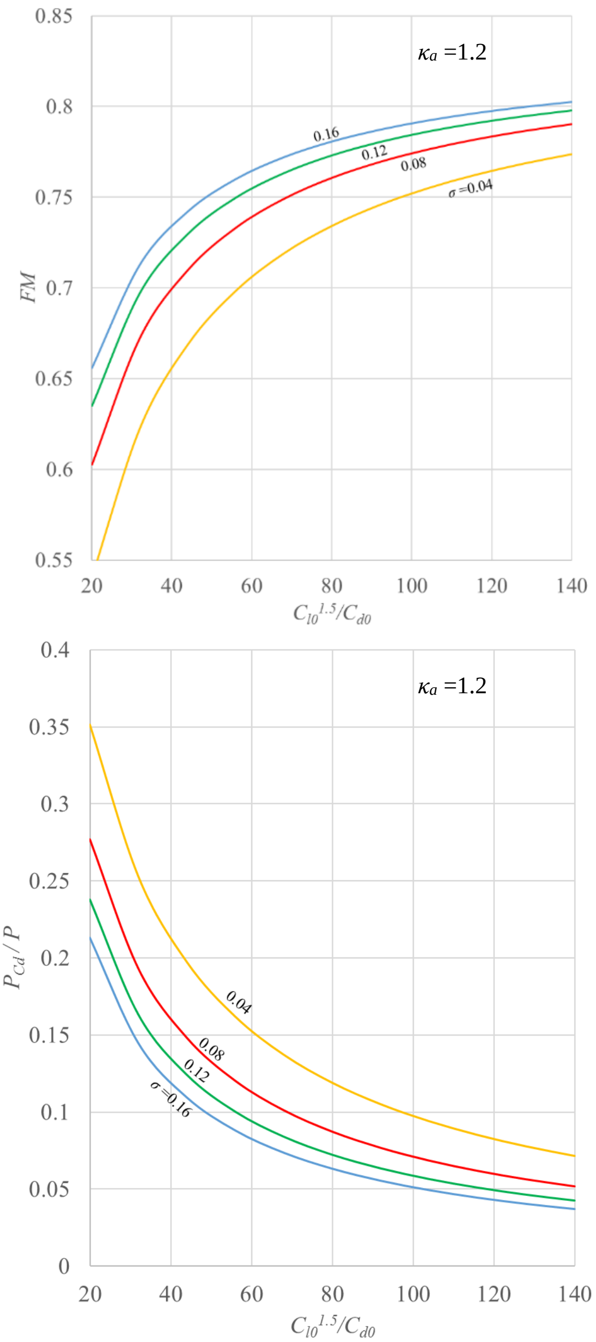

Figure 4, Hover propeller performance estimations

As shown in this chart, influence of the airfoil aerodynamic characteristics is low. It suggests that a ratio of 1:5 up to 1:20 is exists between airfoil-drag and total required power. This means that improvement with the 2-D airfoil aerodynamics is translated to much lower influence on the propeller performance - this emphasizes the low impact of the sectional drag on *FM*.

5. HOVERING PROP. DEMONSTRATION

To demonstrate the above results, a well analyzed propeller is used – Mejzlik 18×6 [23]. It should be emphasized that this is merely a representative example which demonstrates the former, much general results, of Eqs. (17) and (18).

While these equations are general in form, namely true for all propeller's size and operating conditions, the following example is for a given propeller. Still, this demonstration might shed light on the former analytical relations using well analyzed example.

Figure 5 shows the Mejzlik 18×6 propeller and Figure 6 depicts its geometric properties as function of radial coordinate, r, i.e. pitch, β, chord-to-radius ratio, c/R, and thickness-ratio, t/c, distribution. The propeller radius is R= 0.23 m, and for 2 blades configuration, its solidity is σ =0.066 or AF =78 per blade - relatively slender blade.

Table 1, Mejzlik 18×6 performance for T=2.8 kgf

N_b	Σ	Ω rpm	P W	FM	P_{Cd}/P	C_{l0}	$C_{l0}^{1.5}/C_{d0}$
2	0.066	5,010	333	0.68	0.117	0.850	58.5
3	0.099	4,490	348	0.65	0.155	0.704	34.6
4	0.13	4,220	362	0.62	0.188	0.600	23.7

(*) κ_a =1.3 is assumed

Figure 5, Mejzlik 18×6 Propeller, front and top views

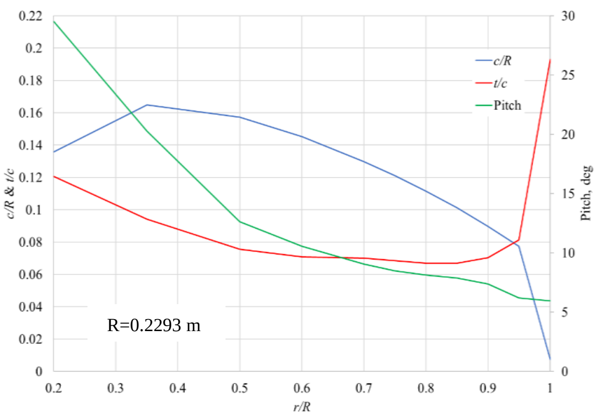

Figure 6, Mejzlik 18×6 Propeller, geometric properties

The propeller in this effort is specified according to its produced thrust. At design conditions, the Mejzlik 18×6 is required to produce T = 2.8 kgf which is established as the required thrust for hover (static operation). All analyses are done for sea-level, ISA.

The same blade geometry is replicated for 2, 3, and 4 bladed propellers, thus range of solidities, σ, are to be explored. Table 1 contains the performance of these propellers, as resulted from BEM analysis. From the resulted thrust and power, the figure-of-merit, *FM*, and P_{Cd}/P are calculated using Eq. (6) and (19).

Figure 7 shows the distribution of Reynolds and Mach number. Reynolds number is Re~100,000 ÷ 150,000. This is quite low, thus the airfoils exhibit relatively high drag, especially at the blade-tip, where the thicknes-ratio is high and Reynolds number is low - this one of the reasons for the relatively low *FM*. Note that the Mejzlik 18x6 propeller is a typical small propeller and is considered adequate for hovering purposes. Hence, the results reflect the situation in most cases of small propellers. For large propellers, where the 2-D characteristics are better, the improvement potential due to airfoil improvement is even lower.

2-D properties of C_{l0}, and $C_{l0}^{1.5}$ / C_{d0}, are depicted in Table 1. Actual sectional distributions of the propeller C_l and C_d are shown in Figure 8. As mentioned in eqs. (9) and (14), C_{l0} and C_{d0} are integral measures of C_l and C_d distribution.

Notice that Table 1 values seem somewhat conservative compared to the actual distribution in Figure 8. It might be due to accumulating influence of following reasons.

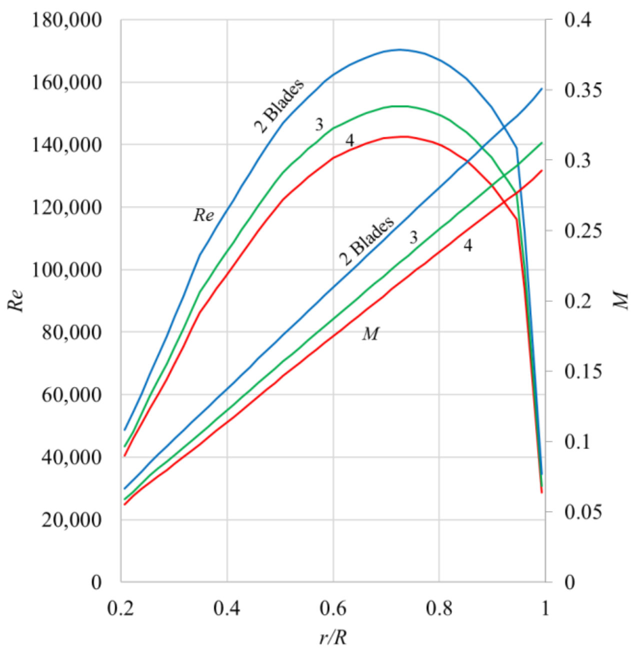

Figure 7, Mejzlik 18×6 Reynolds and Mach number distribution, T=2.8 kgf

a. Actual induced velocity is higher by 30%, hence κ_a ~1.3. Lower chart in Figure 8 shows the actual distribution of κ_a.

b. Tip sections exhibits higher drag-coefficient due to much lower Reynolds number and high thickness-ratio. The drag coefficient is weighted according to r^3 - see Eq. (9), thus the equivalent drag coefficient is higher.

Still, the estimated calculation of the radius averaged C_{l0}, and $C_{l0}^{1.5}/C_{d0}$, as presented in Table 1, fits fairly well to the actual distributions as shown in Figure 8.

To examine potential impact of an improved airfoil, increased and decreased $\Delta C_l^{1.5}/C_d$ are implemented. The results are presented in Table 2.

Although the airfoil aerodynamic, namely $\Delta C_l^{1.5}/C_d$, is changed by ±25%, the resulted power is influenced by merely 3%÷5% while *FM* is changed by about 3%. This is somewhat disappointing considering the large changes with the airfoil aerodynamic.

This demonstrates the limited influence of airfoil aerodynamic characteristics over propeller performance. Note, that for the current case with relatively low Reynold number, airfoil improvement potential is high. For other cases, which are already starting with high *FM*, the influence is expected to be even less pronounce.

Table 2, Mejzlik 18x6 performance influence by $C_l^{1.5}/C_d$, for T=2.8 kgf

N_b	$\Delta C_l^{1.5}/C_d$	Ω, rpm	P,W	ΔP, %	*FM*
2	-25%	5,020	350	5.2	0.645
	0	5,010	333	0	0.679
	+25%	5,000	323	-3.1	0.701
3	-25%	4,500	366	5.2	0.618
	0	4,490	348	0	0.650
	+25%	4,490	337	-3.1	0.671
4	-25%	4,230	382	5.4	0.592
	0	4,220	362	0	0.624
	+25%	4,210	350	-3.2	0.645

6. SUMARRY

This paper focuses with propeller's air-sectional 2-D aerodynamic properties. The air-section selection is based on numerous criteria. It is shown that considering the propeller performance, most crucial criterion is the lift-to drag ratio, or a similar metric based mainly of the cross-sectional drag. Moreover, this metric influences the performance in limited way - its impact is small.

The reason for this low influence is the source of propeller required power. Airfoil drag is one of 3 sources at axial flight and 2 sources at hover. During axial flight most of the required power is due to the propulsive power and during hover most of the required power is due to the induced power. The airfoil drag has low influences on the total required power.

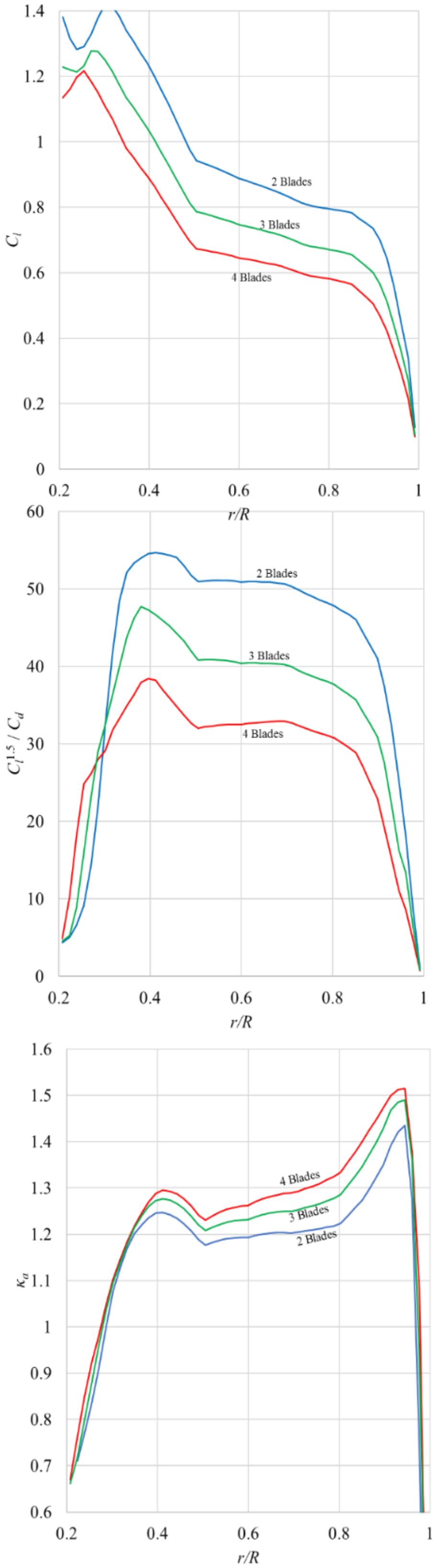

Figure 8, Mejzlik 18×6 sectional aerodynamic properties distribution

For axial flight, changing the drag by 25% influence less than 1% on the propeller efficiency. For the hover performance changing the drag by 25% changes the figure-of-merit by 3% or less. This demonstrates the low impact of airfoil drag characteristics.
One should remember that although propeller's performance is based on lift-to-drag ratio, it is only a single metric out of many others, e.g. thickness for structural and manufactural reasons. As-long-as the airfoil exhibits reasonable drag characteristics, one should focus on these other metrics.
Hence, using optimization to design propeller to best performance by geometric shaping of its air-sections is not relevant. One should choose reasonable existing air-section and focus on the pitch/chord distributions. Thickness distribution, for example, should be considered mainly according to structural constraints.

7. AKNOWLEDGMENT

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